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FGA-1 Fractal Graphene Aggregate Technical Datasheet

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Product Number: FGA-1	Product: Fractal Graphene Aggregate
Product Description	
A few layer, non-functionalized, turbostratic graphene in a powdered form of aggregated nanoplatelets from carbon-rich gas explosion synthesis	

Product Information	
Production Method	Explosion Synthesis
Raw Material	Carbon Bearing Gas
Forms of Materials	PWD – Dry Powder

Characteristic	Test Method	Value
SP2 Bonded Carbon	RAMAN, XPS	Yes (G peak), 100% sp ² (D parameter)
Structural Defects	RAMAN	D/G = 0.68 G width = 39 cm ⁻¹
Number of Layers	X-Ray Diffraction	6-layer average
Z-Axis Dimensions	X-Ray Diffraction	3±0.5nm
Primary Particle Shape	TEM, Light Scattering	Platelets (aggregated)
Lateral Dimensions	TEM	20-50nm
Aspect Ratio	TEM	1:15
Bulk Density	Manual Tapping	70-100 mg/mL*
Chemical/Elemental Analysis	Chemical Analysis	C 99.8%
Oxygen Content %	Chemical Analysis	0.2%
Impurities %	Chemical Analysis	None
Functionalization	Chemical Analysis	Not Detected
Surface Particle Charge	Zeta Potential	~+13mV (pH ~6.7 in DI Water)
Graphene Orientation	RAMAN, XRD	Turbostratic
Specific Surface Area (SSA)	BET	130-180 m ² /g
Crystallinity	Electron Diffraction, X-Ray Diffraction	Crystalline

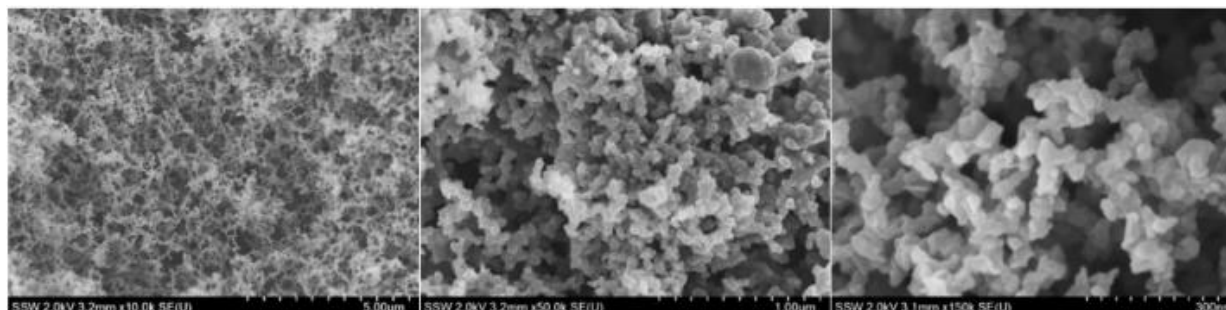
*Density may vary



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Parameters	
Appearance	Black fluffy powder
Number of Layers	≤ 10 layers
Lateral Size	Nano-platelets 20 to 50nm. Aggregates radius of gyration ~150nm
Shape and Form	Fractal aggregate of nano-platelets
Elemental Analysis	Atomic %: 99.8% Carbon, 0.2% Oxygen, No PAHs
Dispersants/Surfactants	None
Concentration	100%
Solid Content	100%
Solvent content	N/A
Substrate Material	N/A
Sheet Resistance	Not applicable
Color	Light absorbing. Black $L^*=2.6$, $a^*=-0.12$, $b^*=-0.79$ (100 observer/D65 Illuminant)
Odor	None
Solubility in Water	Hydrophobic
Electrical Conductivity	Function of powder compression. 100-300 S/m
Thermal Stability	Thermo-gravimetric analysis (TGA) shows: In nitrogen- No volatiles up to 700°C In air- Stable up to 544°C, $T_{max}=725.5^{\circ}\text{C}$ (at $dT/dt=10^{\circ}\text{C}/\text{min}$)

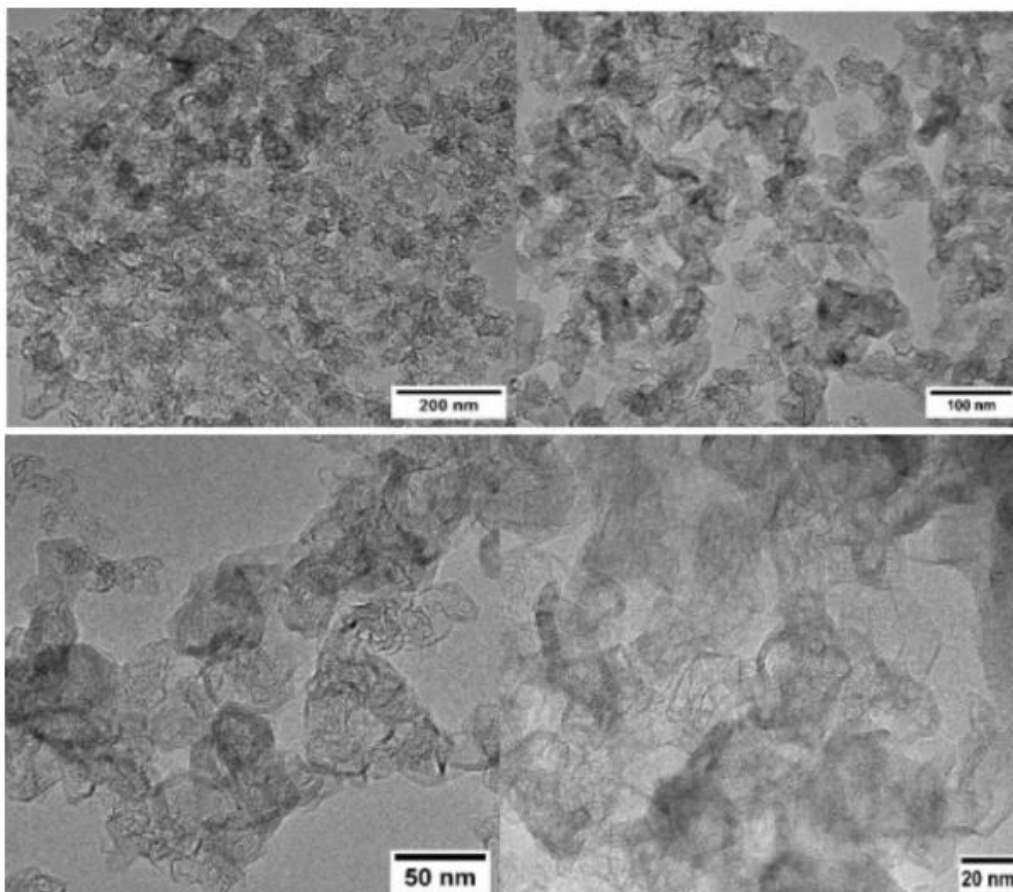
Scanning Electron Micrographs





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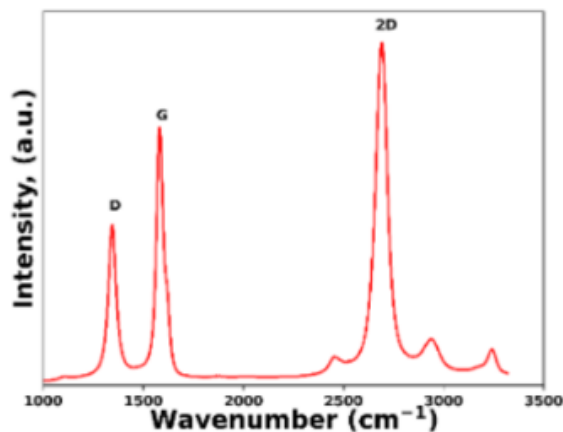
Transmission Electron Micrographs



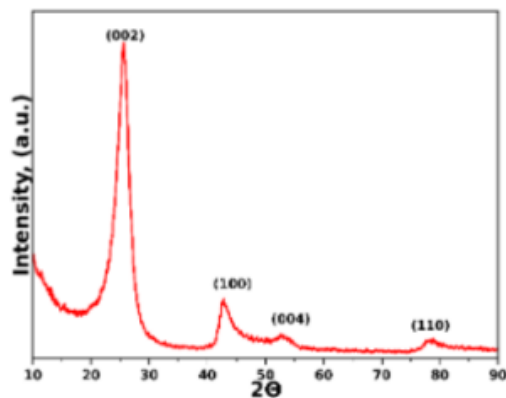


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Raman Spectrum (532 nm)



X-Ray Diffractogram



Notes on Analysis:

- The turbostratic nature is indicated by the asymmetric (100) peak in the X-Ray Diffractogram and the symmetric Lorentzian 2D peak in the Raman Spectrum.
- AFM- The aggregate nature of our fractal graphene is not amenable to AFM analysis.
- Raman- The nanoscale lateral dimensions of our monomer platelets lead to a high fraction of defect edge sites which enhance the intensity of the Raman D bands.

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